

2. Combustion

Why more fuel makes it worse

The single most useful lesson in this project

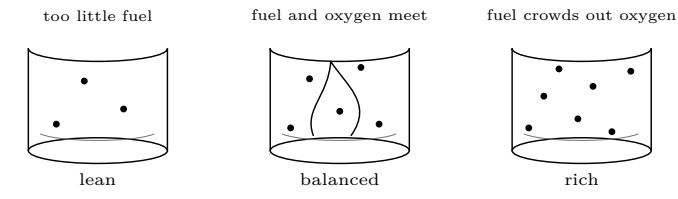
Lesson 2 · about 75 minutes · adult handles all flames and fuel

What you need

- Candles, matches, glass jars of several sizes
- A shallow dish of water
- A tealight, a coin, and a bottle
- Steel wool and a 9 V battery
- Flour or lycopodium powder, a candle, and a tube (outdoors)
- Stopwatch

Predict first

A candle burns under a jar. You use a jar twice the volume. Does it burn twice as long, more than twice, or less? Write a number. Then: a chamber sprayed with twice the fuel — does it fire twice as hard?



Do it

1. **Candle under a jar.** Time how long a candle burns under jars of different sizes. Plot time against volume. It is close to a straight line through the origin — the candle is consuming a fixed *ingredient*, and there is a fixed amount of it per litre of air.
2. **The rising water.** Stand a candle in a dish of water, light it, invert a jar over it. When it goes out, water rises into the jar. Roughly a fifth of the volume, which is the oxygen fraction of air.
3. **Fuel with no air.** Touch a 9 V battery across steel wool. It burns fiercely — iron is a fuel too, given enough surface area.
4. **Too much fuel.** Outdoors, with an adult: puff a small amount of flour through a candle flame and it flashes. Now try to light a *pile* of flour with a match. It will not burn. Same substance, entirely different outcome.
5. **Suffocate one.** Light two candles under identical jars, but put a second lit candle under one of them. Predict, then time.

What it means

Burning needs three things	Fuel, oxygen, and enough heat to start. Remove any one and it stops — which is why the jar candle goes out, why the flour pile will not light, and why you fan the chamber between shots.
Air is one-fifth oxygen	About 21%. The other 79% is mostly nitrogen, which does not burn but does get heated — and heated gas is exactly what pushes your potato, so the nitrogen is doing real work.
The stoichiometric ratio	There is one ratio of fuel to air at which both are used up exactly. Above it the mixture is <i>rich</i> ; below it, <i>lean</i> . Both burn worse than the ideal, and far enough either way it will not ignite at all.
Why more fuel is worse	This is the whole lesson. A chamber with too much hairspray in it has no oxygen left to burn it with. It will not fire, and the next thing most people do — add more — makes it further from working, not closer. <i>see the Safety sheet</i>
The flammability window is limited	A mixture can be too lean or rich. This explains a hazard; it does not yield a launcher dose. Container pressure, nozzle, formulation, and temperature defeat timing by spray count.
Exhaust does not burn	After a shot the chamber is full of carbon dioxide and water vapour. There is no oxygen left. Fanning it out is not superstition — it is the difference between a shot and a misfire.

Break it

Two candles, one jar	Half the time, roughly. Two consumers, one fixed supply.
Surface area	A block of wood is hard to light; sawdust flashes. Powdered flour explodes. The chemistry has not changed — only how much fuel is touching oxygen at once. Hairspray is an aerosol for exactly this reason.
Predict the launcher	Your chamber is about 1.4 L. Roughly a fifth of that is oxygen. Before you ever fire it, argue out how much fuel could possibly be burnt by that much oxygen — and notice how small the answer is.
Never map it by firing	Varying spray count is uncontrolled fuel tuning. It deliberately seeks failures and stronger shots, and is forbidden.

The link to the launcher

Almost every complaint about a combustion launcher — it will not fire, it fires weakly, it fires once and then never again — is a fuel-air ratio problem, and almost always it is too rich rather than too lean. Once a teenager has watched a pile of flour refuse to light while a puff of the same flour flashes, they understand the fix without being told it.

Measure without a launcher

Trial	Fuel arrangement	Oxygen access	Observation / uncertainty
Steel wool	spread / compact	open air	
Candle A	one flame	fixed jar	
Candle B	two flames	same jar	
Flour	dispersed / pile	adult demo	

Questions

Which result distinguishes more fuel from more surface? What evidence shows oxygen became limiting? Why can none of these trials determine a launcher dose?

ACCEPTANCE GATE

Pass only when everyone can explain: no top-up after a misfire, no dose-finding series, and no classroom result authorizes a launcher procedure.

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